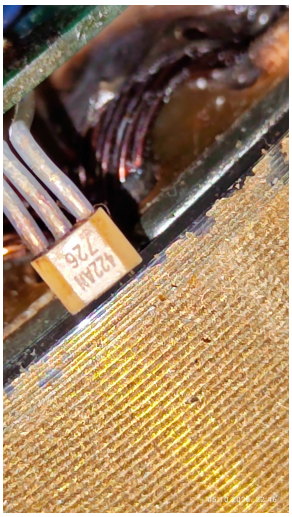
Nilox X4 BLDC hub motor: 422A / 422AN Hall sensor replacement

Field repair note | 15 October 2026
Search terms: 422A, 422AN, 422A vs 422AN, SS41F, Hall polarity, Nilox X4, 60-degree / 120-degree Hall sequence.

Confirmed repair outcome: A 36 V / 250 W geared BLDC hub motor operated correctly with its existing 36 V / 13 A controller after the middle replacement SS41F Hall sensor was installed with its sensing face reversed relative to the two outer SS41F sensors. Its power, ground and output connections were preserved by correctly rerouting the leads.

Original installation

The motor contained three Hall sensors on a curved PCB: black **422A** at each end and a yellow/gold **422AN** in the middle. All three original sensor bodies appeared to face the same way. The original parts were damaged; no verified manufacturer datasheets were available. Additional observed markings included 733 (black) and 726 (yellow).



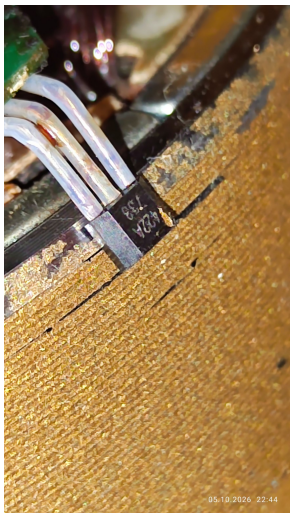


Figure 1. Original sensor markings, photographed before replacement.

Diagnosis and decisive experiment

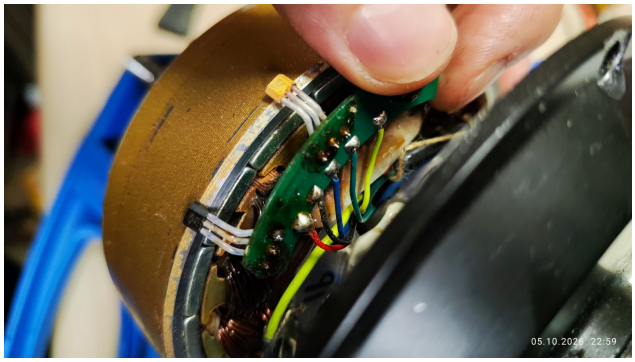
Replacing all three original sensors with identically oriented SS41F units produced buzzing, jerking and unreliable rotation with the existing controller. Individual Hall channels switched approximately between 0 V and 5 V. One manually recorded six-state sequence was 001 → 011 → 111 → 110 → 100 → 000 → 001 (bit order: blue, green, yellow); these readings were preliminary because only one multimeter was available.

Inverting the middle/green bit would yield 011 → 001 → 101 → 100 → 110 → 010, a conventional six-state pattern without 000 and 111. The middle SS41F was therefore physically reversed, with its leads rerouted to maintain correct VCC/GND/OUT. **After this modification, the motor ran normally.**

What the result does — and does not — establish

This is strong practical evidence that the original motor/controller combination requires the middle Hall channel to have the opposite effective magnetic response from three identically facing SS41F units. It is consistent with the original 422AN being magnetically complementary to the two 422A sensors in this motor. It does **not** establish a universal 422A-versus-422AN specification across manufacturers or batches, nor does it prove the original controller design is universally 120°.

Photographic evidence of PCB layout and successful replacement



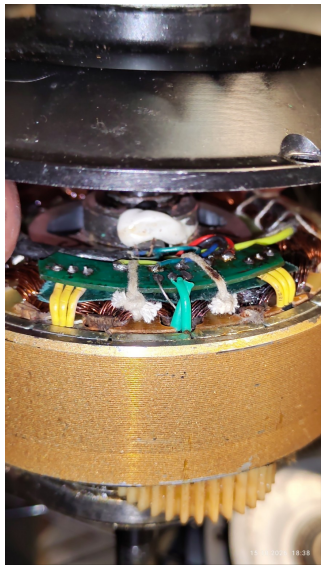


Figure 2. Before and after views of the Hall PCB.

Replacement part identification

The replacement parts were sold as SS41F Hall-effect sensors in a 10-piece pack labeled **41F-TO92-Bao**. The package listing also mentions A3144, SS49E and SS44E; these are not evidence that those types are interchangeable. The working repair specifically used SS41F units.

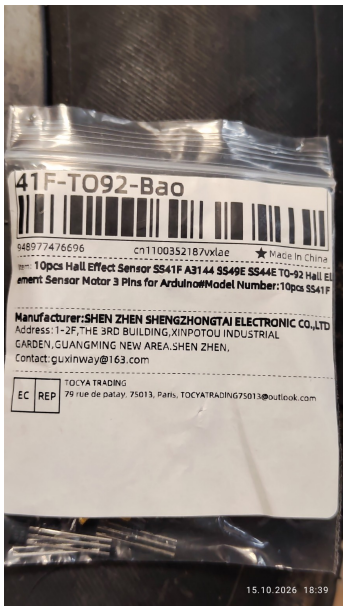


Figure 3. Label on the purchased SS41F sensor pack.

Reproducibility and caution

For a similar motor, first document the exact original positions and orientation of each sensor and check the pinout of the chosen replacements. Reversing a sensor body can reverse its magnetic response, but the leads must be routed so that VCC, GND and OUT remain correctly connected. Verify 0/5 V switching and start with brief unloaded tests and current monitoring. This note documents one successful repair; it does not guarantee that every motor using 422A/422AN markings has the same geometry.

Additional context: The motor windings showed substantial localized heat damage, and the previous repair history is unknown. Successful rotation alone does not rule out winding defects. The controller label and motor markings both suggest May 2018 manufacture, supporting but not proving that the controller is original.